The human heart

The heart is a muscular organ that pumps blood through the body. It has four chambers: the right atrium, the right ventricle, the left atrium and the left ventricle. The sinoatrial node starts each heartbeat; the sinoatrial node is the natural pacemaker. Valves stop blood flowing backwards. The septum separates the left and right sides of the heart.

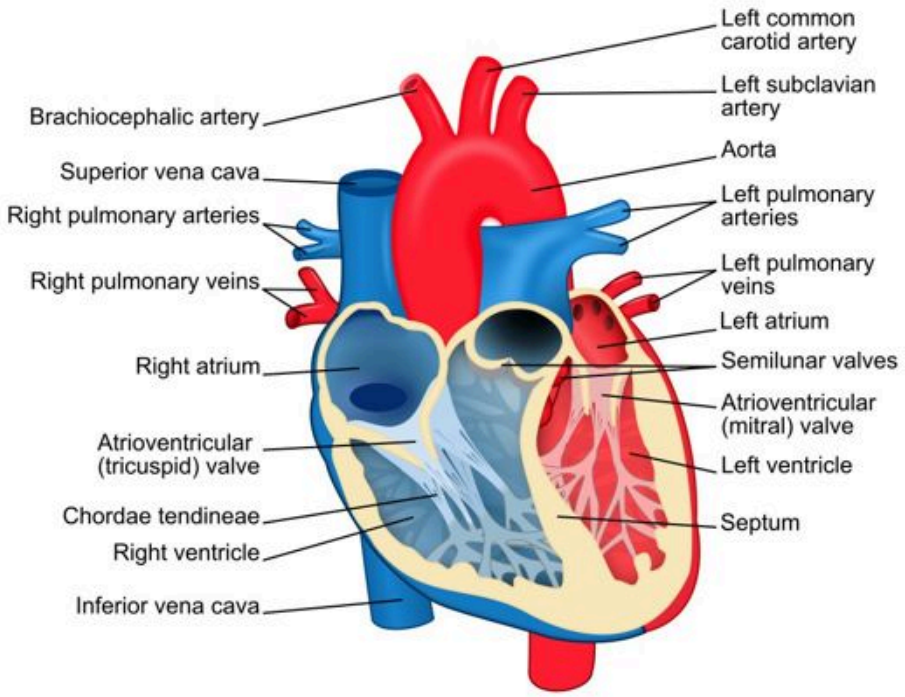


Figure 1: Labelled diagram of the human heart showing the four chambers

The chloroplast

Photosynthesis takes place in the chloroplast. Chlorophyll in the thylakoid membranes absorbs light energy. The light reactions happen in the thylakoid; the Calvin cycle happens in the stroma and fixes carbon dioxide into glucose. Chlorophyll reflects green light, which is why leaves look green. Each thylakoid stack is a granum.

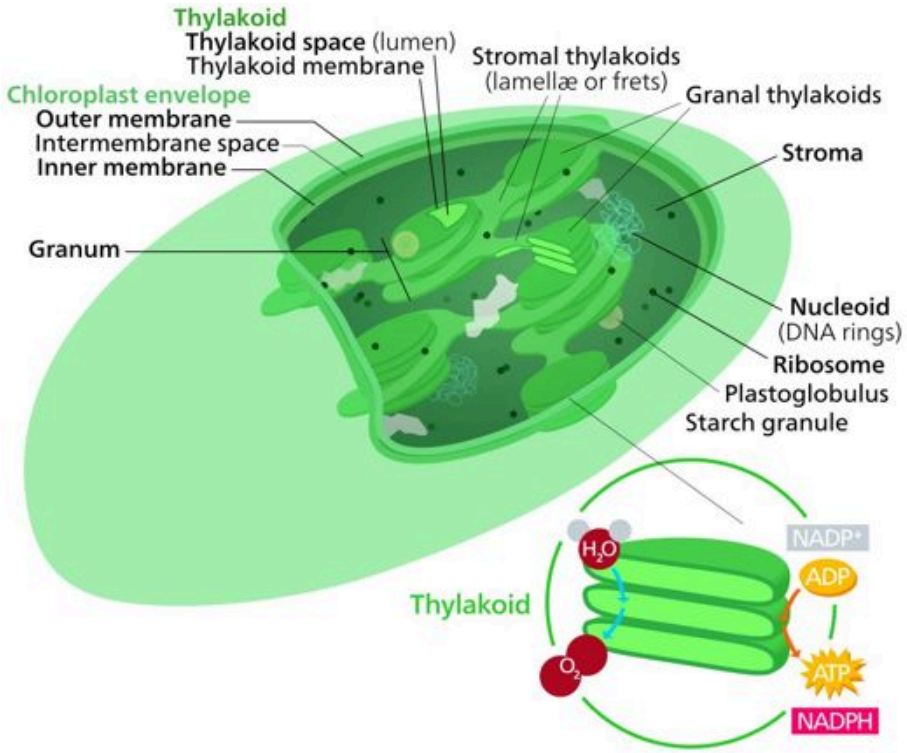


Figure 2: Structure of a chloroplast with thylakoid stacks (grana)

The human skeleton

The adult human skeleton has 206 bones. It supports the body, protects organs such as the brain and the heart, and works with muscles to move the body. Red bone marrow makes blood cells. The femur is the longest bone.



Figure 3: The human skeleton, front view